

Demonstration of Proposed Features

Date	29 June 2026
Team ID	
Project Name	VBCUA
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Speech-to-Text Transcription	Converts uploaded audio into text using OpenAI Whisper.	Implemented	Yes	Core transcription functionality completed successfully.
2	Interactive Streamlit Interface	Provides an intuitive interface for audio upload, topic selection, and result visualization.	Implemented	Yes	Built using Streamlit with custom styling.
3	Semantic Similarity Evaluation	Compares the transcribed explanation with the reference concept using Sentence-BERT.	Implemented	Yes	Generates semantic similarity score for concept understanding.
4	Audio Feature Analysis	Extracts speech metrics such as RMS energy, pause ratio, and filler word ratio using Librosa.	Implemented	Yes	Supports speech fluency and communication analysis.
5	Understanding Score & Report Generation	Calculates the final understanding score, displays qualitative feedback, waveform visualization, and generates a downloadable PDF report.	Implemented	Yes	End-to-end evaluation and reporting demonstrated successfully.
6	Modular Application Integration	Integrates Streamlit UI, Whisper, Sentence-BERT, Librosa, scoring engine, and PDF generation into a unified workflow.	Implemented	Yes	Complete application workflow successfully integrated and demonstrated.

Feature Implementation Summary:

Metric	Value
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Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%